UTSAV PIPERSANIYA

Software Engineer — Python, Django, Distributed Systems
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Available immediately | Open to remote (India), or hybrid/on-site in Pune, Indore, Gurugram

SUMMARY

Backend software engineer with nearly 3 years building production Python systems on Django, PostgreSQL, Redis and Celery. Delivered 100+ production REST endpoints, cut API response times by ~50% through indexing and ORM tuning, and cut long-running workload times by ~60% by moving them onto an asynchronous task fleet. Comfortable at the level where correctness under concurrency actually matters: built a Redis-authoritative bidding engine whose accept path is a single atomic Lua script, with idempotent batched persistence that survives a crash mid-write. Shipped a consumer product end to end, from architecture to live traffic, as a freelance engineer.

TECHNICAL SKILLS

- **Languages** — Python, SQL, JavaScript, TypeScript, Lua (Redis scripting)
- **Backend** — Django, Django REST Framework, FastAPI, Django Channels, Celery, Celery Beat, REST API design, WebSockets, JWT authentication
- **Data** — pandas, NumPy, ETL pipelines
- **Databases** — PostgreSQL, TimescaleDB, query optimization, indexing, ORM tuning, schema design
- **Infrastructure** — Redis, Docker, Linux, Nginx, Gunicorn, Uvicorn, Google Cloud Platform, Azure Blob Storage, Cloudflare, Git, Sentry
- **Frontend** — React, Next.js, React Native (Expo), Tailwind CSS
- **Concepts** — System design, distributed systems, concurrency and atomicity, caching strategy, idempotency, performance optimization, graceful degradation, feature flags

EXPERIENCE

Backend Engineer | Sustainext | Dec 2023 – Jul 2026

Enterprise ESG and sustainability reporting platform.

- Designed and delivered **100+ production REST endpoints** in Django REST Framework, defining server-side contracts directly with frontend teams.
- Cut **API response times by ~50%**, measured before and after, through PostgreSQL indexing, removal of N+1 access patterns, and Django ORM tuning.
- Cut **long-running workload times by ~60%** by moving synchronous processing onto a Celery worker fleet backed by Redis, keeping user-facing APIs responsive under heavy load.
- Built the platform's **caching layer** in Redis, including the invalidation strategy for data that mutates under live reporting workflows.
- Built the **analysis engine** behind ESG reporting: automated emission calculation and aggregation across reporting periods and entities.
- Owned a **fully automated ETL pipeline** end to end — extract from a third-party source, transform to the platform schema with **pandas and NumPy**, load into PostgreSQL — running unattended on a Celery Beat schedule, with live progress streamed over SSE. It made clients' third-party data onboardable and **brought in 5 new clients**.
- Built an in-house emission-factor database that **reduced third-party API costs**.
- Built **Sustainext Chat**, a production AI assistant using retrieval-augmented generation over pgvector with OpenAI models, which also queries live platform data so answers reflect real customer records.
- Built reusable backend services and authentication modules that removed duplicated logic across the codebase; integrated Azure Blob Storage for documents and reports.
- Handled deployment and ongoing maintenance across Docker, Gunicorn, Nginx, Linux and Cloudflare.

Freelance Backend Engineer | Rarita Technologies, Gurugram | Jul 2026 – Present

Collectibles marketplace and live-auction platform.

Rarità — Real-Time Auction Marketplace

Django 5, DRF, Channels, Celery, PostgreSQL, Redis, Docker, GCP.

- Built a **Redis-authoritative bidding engine** that keeps hot auctions off the Postgres write path: authoritative state lives in a Redis hash and a single **atomic Lua script** validates and accepts each bid, so concurrent bids cannot interleave into an inconsistent price.
- Held money as **integer paise** so Lua's double-based arithmetic stays exact, and gave each accepted bid a **monotonic per-listing sequence number**.
- Made persistence **idempotent by sequence number** — a periodic Celery task drains the buffer to PostgreSQL in batches, so a crash between the database commit and the buffer trim can never double-insert. Every auction-close path force-drains before deciding a winner.
- Gated the whole engine behind a feature flag with the original synchronous database path as fallback.
- Built a **time-decayed trending ranking** on hourly Redis hashes, so whole-catalogue recompute costs a fixed number of reads regardless of size, and degrades to lifetime rates rather than failing when Redis is unavailable.
- Implemented WebSocket live bidding with synchronized auction timers via Django Channels, and Celery Beat scheduling for auction activation, completion and notification dispatch.
- Built the full API surface consumed by the React Native client: auth, payments (Razorpay with webhooks), escrow wallet, orders, shipping, notifications.
- Optimized read paths with Redis caching, pagination, indexing, `select_related()` and `prefetch_related()`. Architected for 10,000 concurrent users.

RaritaDrops — live at drops.raritaclub.com

Django + Channels backend; Next.js 15 / React 19 / TanStack Query / Zustand frontend.

- Shipped a live real-time drop-room auction platform end to end — design, implementation, deployment and ongoing maintenance, all owned solo.
- Built instant room creation, bulk item upload, a real-time bidding leaderboard, Buy-It-Now, and a room-end summary with settlement analytics and Excel export.

EDUCATION

B.Tech — Gyan Ganga Institute of Technology and Sciences, 2023 | CGPA 8.34

CERTIFICATIONS

- **Technology Virtual Experience Program** — Deloitte via Forage, 2023
- **Data Analytics Virtual Internship** — Trainity, 2023